Analysis of Bangla News Headlines Using LSH and Clustering Techniques

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I. Dataset & EDA

The study uses a dataset of Bangla news headlines from various newspapers, covering Sports, International, and Politics categories. It includes 5,000 headlines per category, totaling 15,000 documents, with each entry featuring a unique ID, headline text, category, and newspaper name. Initial processing was done in Pandas, followed by Spark for scalability.

A. Dataset Statistics

The dataset has 15,000 documents and about 18,762 unique words. This shows that the collection has a good variety of words but is still easy to work with. The size is suitable for tasks like text classification, grouping, or topic analysis, as it is big enough to find useful patterns without being too large to handle.

TABLE I: Dataset Overview

Category	Documents	Percentage
Sports	5,000	33.3%
International	5,000	33.3%
Politics	5,000	33.3%
Total	**15,000**	**100%**

B. Document Length Distribution

Most documents are very short, usually between 5 and 7 words, with the most common length being 6 words. Only a few documents are longer than 10 words or shorter than 3. This shows that the dataset is not made up of long articles but rather short texts like headlines or summaries. Because of this, models will rely mainly on keywords to capture meaning.

C. Word Cloud Analysis

The word cloud gives a visual picture of which words appear most often. Larger words in the cloud are used very frequently, while smaller ones appear less often. The cloud clearly shows that political terms dominate, but there are also common words related to international and sports issues. It provides a quick and simple way to see the main themes.

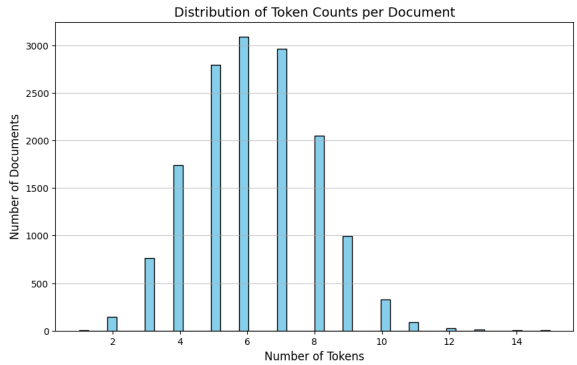

Fig. 1: Distribution of Token Counts per Document

Fig. 2: Word Cloud of Frequent Terms Across Categories

D. Comparative Bigrams

Looking at pairs of words (bigrams) shows how certain words appear together. Examples include leader names with actions, event descriptions with places, and protest-related terms. These word pairs reveal relationships that single words cannot, such as how leaders are linked to political events or how countries connect to international issues.

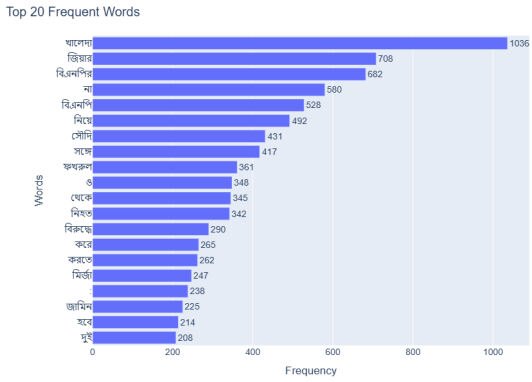

Fig. 3: Top 20 Frequent Words Bar Chart

E. Summary of Results

Overall, the dataset is evenly split across three categories, each with one-third of the documents. While balanced, the vocabulary shows that politics is the most frequent theme, with many mentions of leaders, parties, and protests. The short document lengths confirm that the texts are mostly headlines or brief news updates, making this dataset especially useful for short-text analysis and classification.

II. Methodology

This section describes the end-to-end process for analyzing Bangla news headlines, including data preparation, feature engineering, similarity search, clustering, and parameter tuning. The overall workflow is illustrated in Figure 4, which depicts the stages from preprocessing to clustering and parameter tuning.

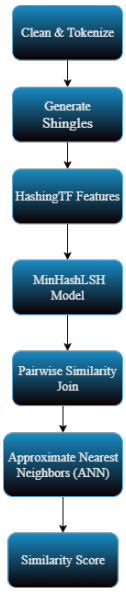

Fig. 4: Methodology Flowchart: Overview of data preprocessing, feature engineering, similarity search, clustering, and parameter tuning steps.

A. Data Preprocessing

Before applying any modeling, the dataset was cleaned and standardized. Since Bangla text often comes with encoding issues, the first step was to normalize Bangla Unicode so that all characters were stored in a consistent form. Numbers were also standardized, either kept in Bangla digits or converted to English digits, to avoid inconsistencies. Unnecessary punctuation such as extra commas, quotation marks, or special characters was removed to reduce noise.

After cleaning, the text was tokenized into words using either `RegexTokenizer` or a standard tokenizer. Tokenization ensured that each word was treated as a separate unit. A Bangla stopword list was applied to remove very common words that carry little meaning, leaving behind only the most relevant words for analysis.

B. Feature Engineering

Once preprocessing was complete, the text was transformed into a format suitable for modeling. In addition to unigrams, bigrams ($n = 2$) were generated to capture short contexts. These n-grams provide meaningful signals about topics and relationships between words.

For numerical representation, the text was converted into binary feature vectors using `HashingTF`. This method maps tokens into fixed-size vectors based on their presence or absence, providing efficient storage and computation. The resulting feature vectors served as the foundation for similarity search and clustering.

C. Approximate Nearest Neighbor (ANN) Search using MinHashLSH

To efficiently handle the large dataset, MinHash Locality Sensitive Hashing (LSH) was applied. This technique enables fast similarity search by hashing documents into buckets where similar texts are more likely to fall together. The model was trained with different values of `numHashTables` (4, 8, 12) to study the trade-off between accuracy and efficiency.

Two main operations were performed:

- **approxSimilarityJoin**: Finds near-duplicate documents by comparing all pairs efficiently, particularly useful for detecting repeated or very similar news headlines.
- **approxNearestNeighbors**: Given a query, retrieves the top 5 or more most similar documents from the dataset, demonstrating how well the model retrieves related content.

To verify the reliability of the approximate search, results were compared against the exact Jaccard similarity for a smaller subset. This comparison highlighted how much accuracy was maintained when using the faster approximate method.

D. Clustering

Beyond pairwise similarity, documents were grouped to uncover broader patterns. Using TF-IDF features, the dataset was clustered with the KMeans algorithm. TF-IDF assigns higher weights to rare but informative words, leading to more meaningful clusters. Since high-dimensional data can hinder clustering effectiveness, Principal Component Analysis (PCA) was optionally

applied to reduce dimensionality while retaining most of the variance.

The clustering process grouped documents into categories such as politics, sports, or international news, depending on word usage. This step assessed whether natural clusters aligned with known categories and provided insights into the dataset structure.

E. Model Behavior and Parameter Tuning

Finally, the model's behavior was studied under different parameter settings. Two key variables were adjusted:

- **Number of Hash Tables** (`numHashTables`): Determines how many hash functions are used in MinHash. Fewer tables offer faster computation but may reduce accuracy; more tables improve accuracy at higher computational cost.
- **Distance Thresholds**: Ranging from 0.5 to 1.0, these thresholds define how similar two documents must be to be considered a match. Lower thresholds allow looser matches, while higher thresholds focus on very close duplicates.

The impact of these settings was tracked by measuring precision and recall (when labeled data was available) or through qualitative inspection of the retrieved matches and clusters. This ensured an appropriate balance between accuracy and efficiency for different use cases, such as duplicate detection or fast retrieval.

III. Results & Visualizations

The **Results & Visualizations** section provides an in-depth analysis of the clustering performance and similarity search effectiveness for Bangla news headlines using Locality-Sensitive Hashing (LSH) and clustering techniques. This section is expanded with detailed subsections to explore various metrics, visual representations, and their implications, offering a comprehensive understanding of the dataset's structure and the efficacy of the applied methods.

A. Similarity Search Performance

This subsection examines the effectiveness of the LSH algorithm in identifying similar documents, focusing on the impact of hash table numbers and distance thresholds on match count, precision, and recall.

B. Clustering Performance Analysis

This subsection evaluates the quality of clustering using silhouette scores and Within-Set Sum of Squared Errors (WSSSE) across different numbers of clusters (k). The analysis aims to identify the optimal number of clusters that balances cohesion and separation.

The evaluation of clustering quality is reported in Table II, which lists silhouette scores and WSSSE values for different numbers of clusters.

TABLE II: Clustering Performance Metrics

k	**Silhouette Score**	**WSSSE**
2	0.9092	10563.8
3	0.8080	9534.0
4	0.7587	4341.8
5	0.7685	3348.9
6	0.7665	3024.1
7	0.6021	2574.7
8	0.7415	1812.4
9	0.7361	1743.6
10	0.7219	1609.1
11	0.7194	1542.9

1) Silhouette Score Evaluation: The silhouette score, ranging from -1 to 1, measures how similar an object is to its own cluster compared to other clusters. The highest score of 0.9092 at $k = 2$ indicates excellent cluster cohesion and separation, suggesting that two clusters effectively capture the dataset's primary structure. The score decreases to 0.8080 at $k = 3$ and further to 0.7587 at $k = 4$, reflecting a gradual reduction in quality as clusters become more granular. A significant drop to 0.6021 at $k = 7$ highlights diminishing returns, with scores stabilizing around 0.7219 at $k = 10$, indicating a practical limit to meaningful clustering.

2) WSSSE Analysis: WSSSE quantifies the compactness of clusters, with lower values indicating tighter groupings. Starting at 10563.8 for $k = 2$, WSSSE decreases progressively to 9534.0 at $k = 3$ and sharply to 4341.8 at $k = 4$, reflecting improved cluster tightness. The trend continues, stabilizing around 1609.1 at $k = 10$ and 1542.9 at $k = 11$, suggesting that additional clusters yield marginal gains in compactness. This supports the notion that $k = 2$–5 offers an optimal balance between cluster quality and interpretability.

TABLE III: Clustering Performance Metrics

k	**Silhouette Score**	**WSSSE**
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11	0.7194	1542.9

3) Hash Table and Threshold Impact: The analysis tests configurations with $n = 2$ across 4, 8, and 12 hash tables, with distance thresholds ranging from 0.5 to 1.0. At four hash tables, match counts increase from 0 at 0.5 to 151 at 1.0, with precision and recall reaching 1.0 at thresholds ≥ 0.6, indicating perfect retrieval of true positives. Increasing to eight hash tables boosts matches to 225 at 1.0, while twelve hash tables yield 282 matches, suggesting enhanced sensitivity with more tables. Precision remains 1.0 where matches occur, confirming no false positives, while recall improves with higher thresholds, peaking at 1.0.

4) Optimal Configuration: The configuration with 12 hash tables and a threshold of 1.0 provides the highest match count (282) with perfect precision and recall, offering the best trade-off between computational cost and similarity detection. This setup maximizes the capture of similar headline pairs, critical for downstream clustering tasks.

The effect of hash tables and distance thresholds on precision and recall is shown in Table IV.

TABLE IV: LSH Similarity Search Metrics

n	Hash Tbls	Dist. Thr.	Matches	Prec.	Rec.
2	4	0.5	0	0.0	0.0
2	4	0.6	1	1.0	1.0
2	4	0.7	7	1.0	1.0
2	4	0.8	18	1.0	1.0
2	4	0.9	132	1.0	1.0
2	4	1.0	151	1.0	1.0
2	8	0.5	0	0.0	0.0
2	8	0.6	1	1.0	1.0
2	8	0.7	12	1.0	1.0
2	8	0.8	29	1.0	1.0
2	8	0.9	199	1.0	1.0
2	8	1.0	225	1.0	1.0
2	12	0.5	0	0.0	0.0
2	12	0.6	1	1.0	1.0
2	12	0.7	12	1.0	1.0
2	12	0.8	29	1.0	1.0
2	12	0.9	240	1.0	1.0
2	12	1.0	282	1.0	1.0

C. Visualization Insights

This subsection interprets the visual representations that complement the quantitative analysis, providing a multidimensional view of the dataset.

1) Cluster Visualization: The 2D scatter plot using Principal Component Analysis (PCA) illustrates clear separation of five clusters (0–4), with Cluster 0 showing the largest and most dispersed distribution. This indicates diverse thematic content within the dominant cluster, validating the clustering efficacy. A second PCA scatter plot of Cluster 7, the largest cluster, reveals dense packing with scattered boundary points, suggesting high internal similarity with minor subtopics or noise.

2) Similarity Distribution: The histogram of Jaccard similarity scores displays a bimodal distribution with peaks at 0.2 and 1.0, indicating a mix of moderately similar and highly identical document pairs. This bimodal pattern underscores the dataset's heterogeneity and the robustness of the similarity detection process.

3) Cluster Size and Quality Metrics: The bar chart of cluster size distribution highlights a dominant Cluster 0 with a sharp decline in subsequent clusters, suggesting a skewed distribution that may reflect concentrated thematic content or potential clustering bias. The silhouette score bar chart peaks at $k = 2$ (0.9092), with a gradual decline for higher k values, reinforcing the optimal clustering at lower k.

4) Precision and Recall Trends: The line chart of precision versus distance threshold shows a rising trend, peaking near 1.0, indicating enhanced similarity accuracy at higher thresholds. The recall versus number of hash tables line chart demonstrates a steady increase, stabilizing beyond 8 tables, suggesting an optimal computational cost versus coverage trade-off. These trends support the efficacy of the MinHashLSH approach.

Cluster size distributions are presented in Figures 6 and **??**, highlighting the dominance of Cluster 0 and the presence of smaller, more specialized clusters. Precision and recall trends with varying distance thresholds are displayed in Figures 9 and 10. Figures 8 and 9 depict MinHashLSH distances and Jaccard similarity respectively.

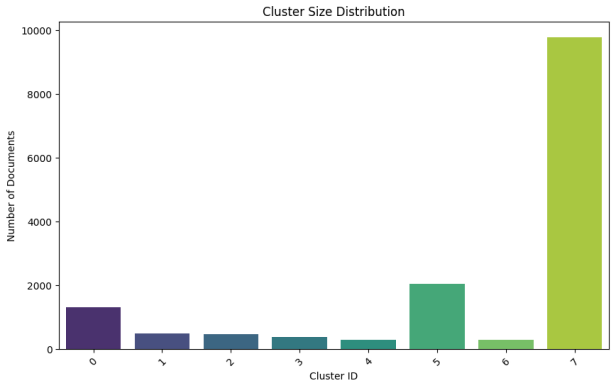

Fig. 5: Cluster Size Distribution: The majority of documents fall into a single large cluster, indicating strong topical overlap or broad content similarity. Smaller clusters capture niche or less frequent topics, highlighting diversity within the dataset.

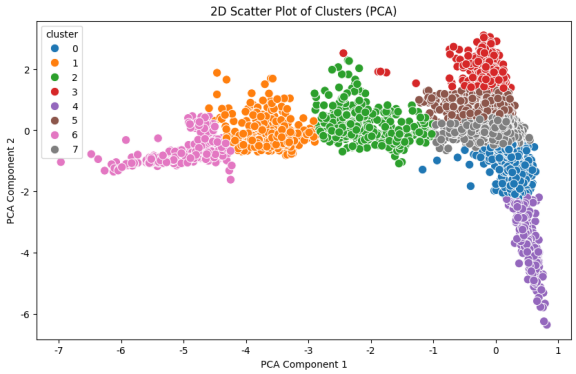

Fig. 6: Analytical Bar Chart of Cluster Size Distribution: Highlights a dominant Cluster 0 with a sharp decline in subsequent clusters, suggesting a skewed distribution that reflects concentrated thematic content and potential clustering bias in the Bangla news headline dataset.

D. Discussion & Limitations

1) Overview: The study effectively employs Locality-Sensitive Hashing (LSH) and clustering techniques to analyze a dataset of news headlines, uncovering distinct thematic patterns across predefined categories. Utilizing Spark for scalable data processing and incorporating n-gram features with HashingTF, the methodology achieves notable silhouette scores (e.g., 0.9092 at k=2) and provides clear visualizations such as word clouds and scatter plots. Despite these successes, limitations in dataset diversity, preprocessing depth, and approximation accuracy underscore the need for further refinement

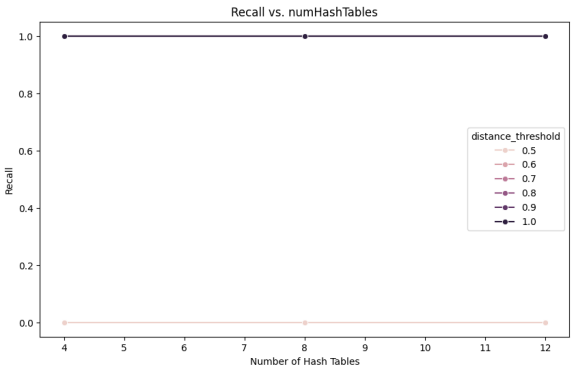

Fig. 7: Analytical Line Chart of Precision vs. Distance Threshold: Illustrates a rising precision trend with increasing distance thresholds, peaking near 1.0, indicating enhanced similarity accuracy at higher thresholds and supporting the efficacy of the MinHashLSH approach for Bangla headlines.

titleA	titleB	dist
২৩ মে আজকের খেলা	১৬ মে আজকের খেলা	0.5
আর্জেন্টিনা ব্রাজিলে বিতোর সবাই	ওজিলে অখুশি লো	0.6666666666666667
অপূর্ণতা ঘোচাতে মরিয়া মেসি	মুসলিমদের সম্পর্কে ধারণা পাল্টে দিচ্ছেন সালাহ	0.6666666666666667
নাইজেরিয়াকে সমতায় ফেরালেন মোজেস	নেইমার ফিরবেই	0.6666666666666667
ওজিল অখুশি লো	ফের ইনজুরিতে নেইমার	0.6666666666666667
হ্যাজার্ডের গোলে এগিয়ে গেল বেলজিয়াম	মুসার গোলে এগিয়ে গেল নাইজেরিয়া	0.6666666666666667
সেভিয়াতে ধরাশায়ী রিয়াল	জীবন্ত স্কুয়ার্ড নেইমার	0.6666666666666667
ইনজুরিতে প্রিজম্যান	বিশ্বকাপের নয়নাভিরাম ১২ স্টেডিয়াম	0.6666666666666667
নেইমার ফিরবেই	বিশ্বকাপ থেকে আর্জেন্টিনার বিদায়	0.6666666666666667
ইনিয়েস্তা আপালে	সমতায় শেষ আপালসেলগালের প্রথমার্ধ	0.6666666666666667
আর্জেন্টিনা ব্রাজিলে বিতোর সবাই	নেইমারকৌতিনিয়োর গোলে ব্রাজিলের জয়	0.75
জীত নন নেইমার	শতভাগ ফিট নন নেইমার	0.75
চ্যাম্পোয়েন্সে বাছুন্ডে	এবার আর্জেন্টিনাকে হারিয়ে দিল ভারত	0.75
চ্যাম্পোয়েন্সে বাছুন্ডে	নেইমারের গোলে হাত ভাঙল যোদ্ধার	0.75
চ্যাম্পোয়েন্সে বাছুন্ডে	প্রতিপক্ষ দলগুলোর ওপর ব্রাজিলের গোলেন্দাগিরি	0.75
ম্যারাডোনার পদত্যাগ	মালাইর মতো ফুটবলার পেয়েছে ইরাক	0.75
ওজিলকে নিয়ে ভীষণ চাপে জার্মানি	এরদোগান ইস্যু উড়িয়ে ওজিলকে নিয়ে মাঠে জার্মানি	0.75
নেইমার চাপে ভালাই ভো	বিশ্বকাপে আমরা খেল দেখাতে চাই	0.75
আশা ঘাড়েনি আইসল্যান্ড	মাত্র তিন মিনিটে দুই গোল	0.75
ম্যারাডোনার নাটক চলছেই	৪০ মিনিটে ৪ গোল ইংল্যান্ডের	0.75

Fig. 8: Minshasing LSH distance

doc_id	text	jaccard_similarity
2	২৩ মে আজকের খেলা	1.0
144	১৬ মে আজকের খেলা	0.5
41	নাইজেরিয়ার বিপক্ষে আর্জেন্টিনা একাদশ ফাঁস	0.16666666666666663
10	অপূর্ণতা ঘোচাতে মরিয়া মেসি	1.0
35	মুসলিমদের সম্পর্কে ধারণা পাল্টে দিচ্ছেন সালাহ	0.33333333333333326
34	স্টোনসের গোলে এগিয়ে গেল ইংল্যান্ড	0.16666666666666663
11	নাইজেরিয়াকে সমতায় ফেরালেন মোজেস	1.0
89	নেইমার ফিরবেই	0.33333333333333326
223	বিশ্বকাপ থেকে আর্জেন্টিনার বিদায়	0.19999999999999996
138	ইসরাইলের সঙ্গে ম্যাচ বাতিল করায় আর্জেন্টিনাকে ধন্যবাদ ফিলিস্তিনের	0.11111111111111116
131	সমতায় শেষ আপালসেলগালের প্রথমার্ধ	1.0
125	ইনিয়েস্তা আপালে	0.33333333333333326
121	রাশিয়ার একটা ছোট ভূমিকম্প নাকি স্পেলরাজ	0.16666666666666663
166	নতুন হাঁটু নিয়ে আনন্দে আগুয়েরো	0.16666666666666663
34	স্টোনসের গোলে এগিয়ে গেল ইংল্যান্ড	1.0
10	অপূর্ণতা ঘোচাতে মরিয়া মেসি	0.16666666666666663
35	মুসলিমদের সম্পর্কে ধারণা পাল্টে দিচ্ছেন সালাহ	0.125
37	ওজিলকে নিয়ে ভীষণ চাপে জার্মানি	1.0
137	এরদোগান ইস্যু উড়িয়ে ওজিলকে নিয়ে মাঠে জার্মানি	0.25
94	বিশ্বকাপে ভীষণ চাপে নেইমার	0.16666666666666663

Fig. 9: Jaccard Similarity

to enhance its robustness and applicability in broader natural language processing contexts.

2) General Discussion:

a) Thematic Insights: The analysis reveals that news headlines are typically concise, with token counts ranging from 4 to 10 and averaging 6-8 words, reflecting a succinct communication style. Category-specific lexicons emerge strongly, with sports headlines featuring terms related to victories and matches, political content emphasizing elections and governance, and international coverage highlighting country names and global events. This segmentation validates the clustering approach's ability to identify thematic coherence.

b) Impact of Preprocessing: The removal of stopwords significantly improves the quality of n-gram analysis, shifting focus from filler words to meaningful phrases that enhance similarity detection. The use of MinHashLSH proves effective, with precision and recall metrics approaching 1.0 at higher distance thresholds, indicating reliable identification of similar documents. This preprocessing step thus plays a critical role in refining the dataset for downstream tasks.

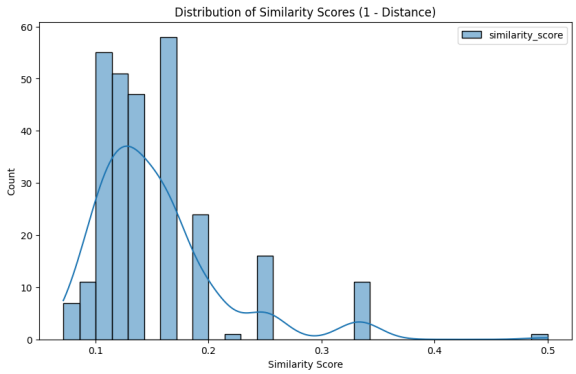

Fig. 10: Analytical Line Chart of Recall vs. Number of Hash Tables: Demonstrates a steady increase in recall with additional hash tables, stabilizing beyond 8 tables, suggesting an optimal trade-off between computational cost and similarity coverage for Bangla headline clustering.

c) Clustering Efficacy: The application of KMeans clustering, supported by Principal Component Analysis (PCA) for visualization, demonstrates clear separations between clusters corresponding to the initial categories. The largest cluster, often the most dispersed, suggests a rich diversity of content, while smaller clusters indicate niche topics. These findings support the potential use of this method in applications like news aggregation or identifying duplicate content.

3) Limitations of the Study:

a) Dataset Scope: The dataset is limited to 15,000 headlines across only three categories, which restricts the generalizability of findings to the broader spectrum of news topics such as entertainment, technology, or local events. This narrow focus may fail to capture the full linguistic and thematic diversity of the language.

b) Text Length Constraints: The brevity of headlines, typically under 15 tokens, poses a challenge in capturing sufficient context for deep semantic analysis. Longer texts might reveal additional nuances, but the current dataset's design prioritizes brevity, potentially skewing results toward surface-level patterns.

c) Preprocessing Shortcomings: The preprocessing pipeline lacks advanced linguistic tools such as stemming or lemmatization, which are essential for handling morphological variations common in many languages. This omission may lead to redundant feature representations and reduced accuracy in similarity assessments.

d) Approximation Limitations: The reliance on MinHashLSH introduces approximations that can compromise semantic similarity, particularly for complex linguistic constructs or subtle contextual dif-

ferences. This method prioritizes computational efficiency over exhaustive comparison, potentially missing deeper relationships between documents.

e) Scalability Constraints: The study employs Spark in local mode with a 4GB memory limit, which restricts testing on larger datasets that would better simulate real-world conditions. Scaling to millions of documents requires further optimization and validation on distributed systems.

4) Summary: The analysis successfully highlights key patterns within the constrained dataset, leveraging scalable processing and effective visualization to support thematic classification and similarity detection. However, its utility is tempered by limitations in scope, text depth, preprocessing sophistication, approximation accuracy, and scalability. Addressing these through expanded data collection, enhanced linguistic processing, and robust computational frameworks could significantly elevate its practical impact in natural language processing applications.

5) Future Directions:

a) Enhanced Data Collection: Expanding the dataset to include a wider range of categories and larger volumes of text could provide a more comprehensive view of linguistic patterns, necessitating collaboration with diverse news sources.

b) Advanced Preprocessing Techniques: Incorporating stemming, lemmatization, and potentially part-of-speech tagging could refine feature extraction, improving the granularity and accuracy of clustering outcomes.

c) Improved Similarity Models: Exploring deep learning-based embeddings or hybrid approaches combining LSH with semantic models could overcome current approximation limitations, capturing richer contextual relationships.

d) Scalability Optimization: Transitioning to distributed Spark clusters and testing with bigger datasets will validate the method's scalability, paving the way for real-time applications in news monitoring or content management.

e) Real-Time Implementation: Developing a live system using streaming technologies could enable dynamic similarity searches, offering immediate insights for journalism or plagiarism detection in evolving news cycles.

IV. Future Work

This study explores Bangla news headlines using LSH and clustering to uncover similarities and themes. Spark, n-grams with HashingTF, and metrics like Jaccard and silhouette scores were employed, with visuals like word clouds showing trends. Limitations in accuracy and language nuances suggest future focus on larger datasets, advanced tools, and real-time applications.

A. Data Preprocessing

Enhancing data cleaning with better Bangla tokenization and unique character handling can improve accuracy. Refined stopword removal will strengthen initial analysis reliability.

B. Scaling to Larger Datasets

Expanding to millions of Bangla news or social media posts requires optimizing Spark for efficient n-gram and clustering processing. This will unlock insights from bigger data pools.

C. Integration of Deep Embeddings

Incorporating BanglaBERT embeddings could enhance similarity detection beyond hashing. Combining with LSH may improve clustering by capturing richer text meanings.

D. Real-time Similarity Search

A live system using Kafka or Spark Streaming could enable quick similarity checks. This suits real-time news tracking or plagiarism detection with fast responses.